Titanicoct26

October 31, 2026

[2]:
```
# Importar las librerias necesarias
import pandas as pd
import matplotlib.pyplot as plt
```

[3]:
```
# Cargar mis datos
data = pd.read_csv("2titanic.csv")
```

[4]:
```
# Leer los primeros 5 renglones
data.head()
```

[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th…	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

[5]:
```
# Para saber cuantas columnas y renglones tiene mi dataset
data.shape
```

[5]:
```
(891, 12)
```

[6]:
```
# Cuántos renglones tiene cada columa
data.count()
```

[6]:
```
PassengerId    891
Survived       891
Pclass         891
Name           891
Sex            891
Age            714
SibSp          891
Parch          891
Ticket         891
Fare           891
Cabin          204
Embarked       889
dtype: int64
```

[7]:
```
col_names = data.columns.tolist() # Convierte los nombres de las columnas en una lista y las guarda en la variable col_names
col_names
```

[7]:
```
['PassengerId',
 'Survived',
 'Pclass',
 'Name',
 'Sex',
 'Age',
 'SibSp',
 'Parch',
 'Ticket',
 'Fare',
 'Cabin',
 'Embarked']
```

[9]:
```
for col in col_names:
    print("Valores nulos en", col, ":", data[col].isnull().sum()) # Esta funcion hace una sumatoria de todos los valores nulos por columna
```

```
Valores nulos en PassengerId : 0
Valores nulos en Survived : 0
Valores nulos en Pclass : 0
Valores nulos en Name : 0
Valores nulos en Sex : 0
Valores nulos en Age : 177
Valores nulos en SibSp : 0
Valores nulos en Parch : 0
Valores nulos en Ticket : 0
Valores nulos en Fare : 0
Valores nulos en Cabin : 687
Valores nulos en Embarked : 2
```

[10]:
```
# Vamos a ver la estadística descriptiva de mi dataset
data.describe()
```

[10]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

[11]:
```
data.head()
```

[11]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th…	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

[12]:
```
def sex_transform(x):
    d = {'male':'M','female':'F'}
    return(d[x])

data["Sex"] = data['Sex'].apply(sex_transform)
```

[13]:
```
data.head()
```

[13]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	M	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th…	F	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	F	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	F	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	M	35.0	0	0	373450	8.0500	NaN	S

[14]:
```
# Cuántos y cuáles pasajeros se subieron sin pagar?

data[data.Fare == 0] # Muestrame todos los renglones que cumplan la condicion
 ↪de que la columna fare sea igual a 0
```

[14]:

	PassengerId	Survived	Pclass	Name	Sex
179	180	0	3	Leonard, Mr. Lionel	M
263	264	0	1	Harrison, Mr. William	M
271	272	1	3	Tornquist, Mr. William Henry	M
277	278	0	2	Parkes, Mr. Francis "Frank"	M
302	303	0	3	Johnson, Mr. William Cahoone Jr	M
413	414	0	2	Cunningham, Mr. Alfred Fleming	M
466	467	0	2	Campbell, Mr. William	M
481	482	0	2	Frost, Mr. Anthony Wood "Archie"	M
597	598	0	3	Johnson, Mr. Alfred	M
633	634	0	1	Parr, Mr. William Henry Marsh	M
674	675	0	2	Watson, Mr. Ennis Hastings	M
732	733	0	2	Knight, Mr. Robert J	M

```
806          807         0       1            Andrews, Mr. Thomas Jr   M
815          816         0       1                 Fry, Mr. Richard   M
822          823         0       1   Reuchlin, Jonkheer. John George   M

      Age  SibSp  Parch  Ticket  Fare Cabin Embarked
179  36.0      0      0    LINE   0.0   NaN        S
263  40.0      0      0  112059   0.0   B94        S
271  25.0      0      0    LINE   0.0   NaN        S
277   NaN      0      0  239853   0.0   NaN        S
302  19.0      0      0    LINE   0.0   NaN        S
413   NaN      0      0  239853   0.0   NaN        S
466   NaN      0      0  239853   0.0   NaN        S
481   NaN      0      0  239854   0.0   NaN        S
597  49.0      0      0    LINE   0.0   NaN        S
633   NaN      0      0  112052   0.0   NaN        S
674   NaN      0      0  239856   0.0   NaN        S
732   NaN      0      0  239855   0.0   NaN        S
806  39.0      0      0  112050   0.0   A36        S
815   NaN      0      0  112058   0.0  B102        S
822  38.0      0      0   19972   0.0   NaN        S
```

[15]:
```
# Tabla cruzada entre supervivencia y sexo
pd.crosstab(data.Survived, data.Sex)
```

[15]:
```
Sex         F    M
Survived
0          81  468
1         233  109
```

[16]:
```
# Agrupar los datos por clase y sexo, despues solo mostrar la suma de los que␣
 ↪sobrevivieron
pclass_sex_survival_count = data.groupby(['Pclass','Sex'])['Survived'].sum()
pclass_sex_survival_count
```

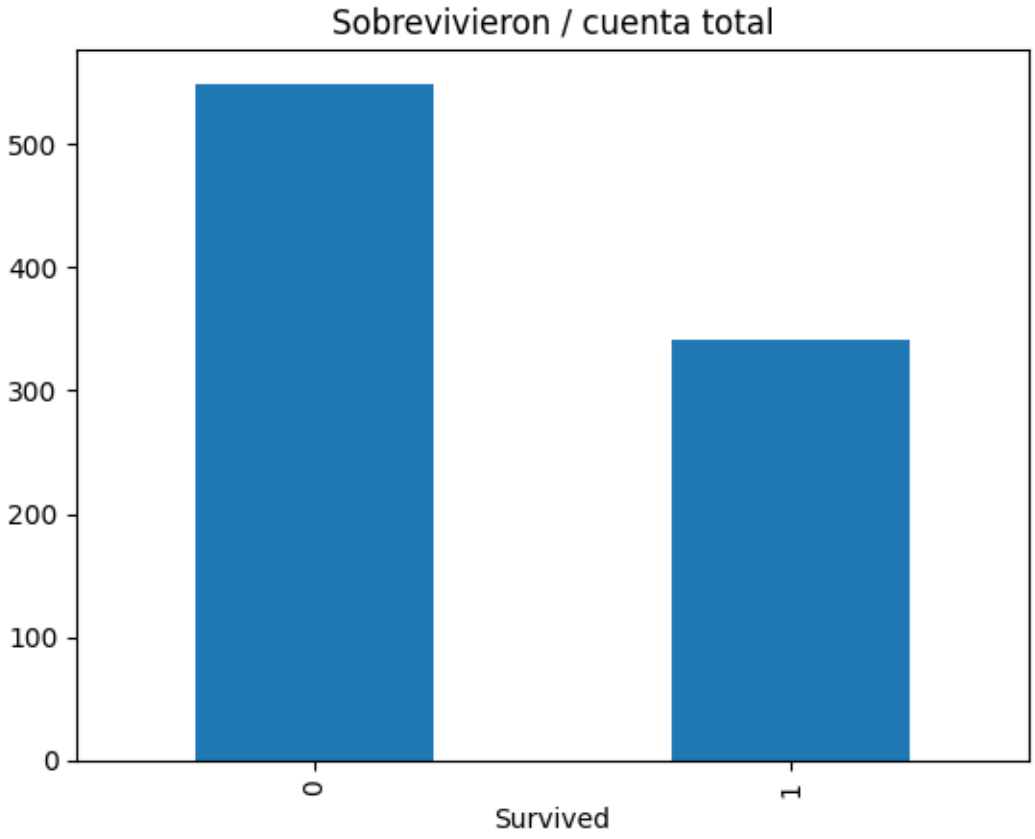
[16]:
```
Pclass  Sex
1       F      91
        M      45
2       F      70
        M      17
3       F      72
        M      47
Name: Survived, dtype: int64
```

[17]:
```
# Tabla cruzada entre Pclass y sexo
pd.crosstab(data.Pclass, data.Sex)
```

[17]:
```
Sex       F    M
Pclass
1        94  122
2        76  108
3       144  347
```

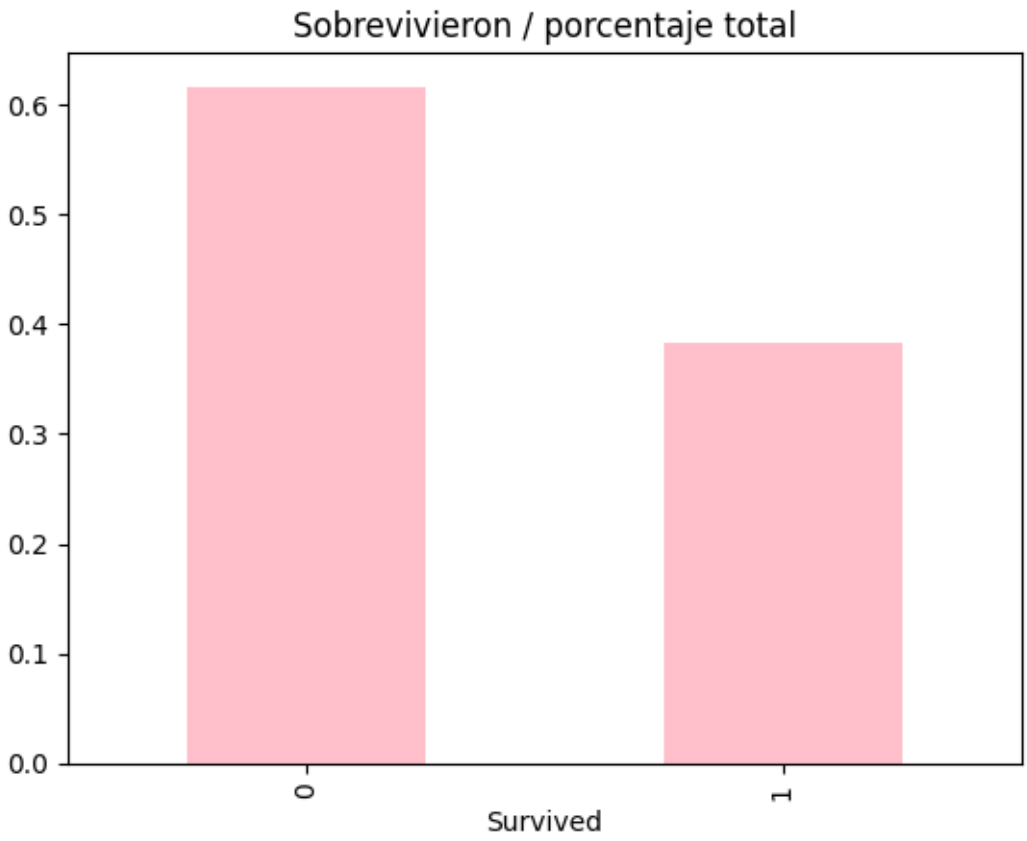
[19]:
```
# Gráfica de la columna survived
data.Survived.value_counts().plot(kind = 'bar')
plt.title("Sobrevivieron / cuenta total")
```

[19]:
```
Text(0.5, 1.0, 'Sobrevivieron / cuenta total')
```



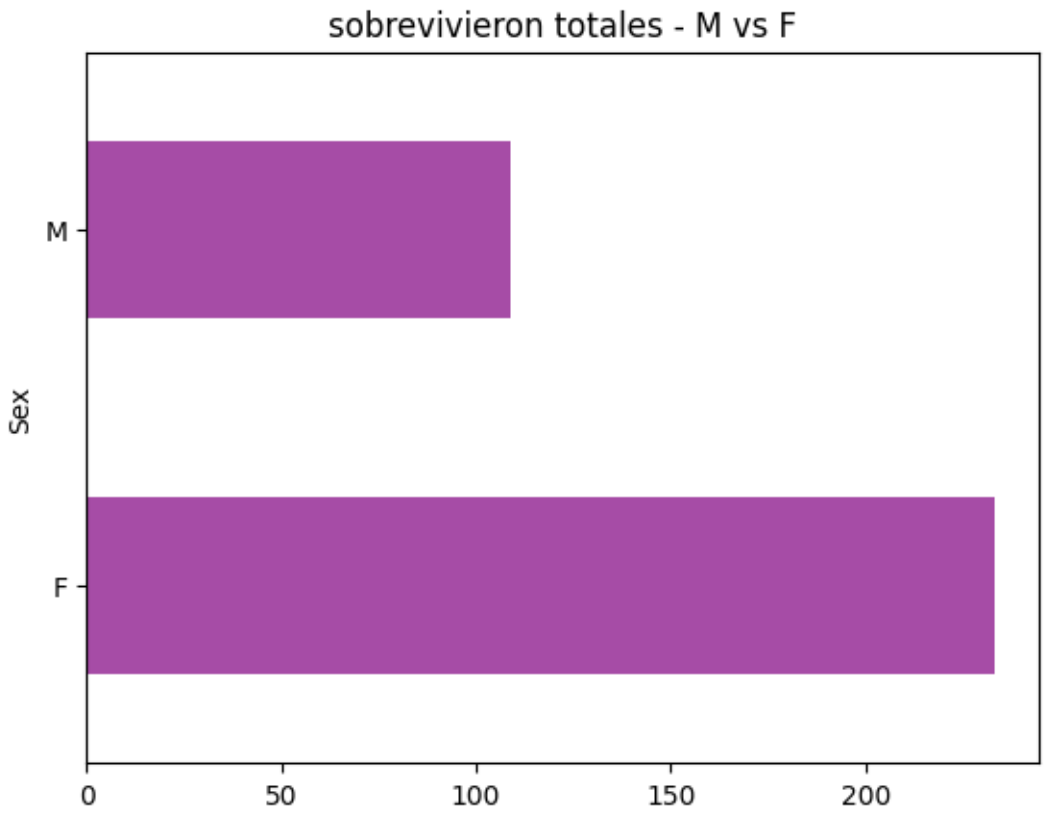
[20]:
```
# Gráfica de la columna survived
data.Survived.value_counts(normalize = True).plot(kind = 'bar', color = "pink")
plt.title("Sobrevivieron / porcentaje total")
```

[20]:
```
Text(0.5, 1.0, 'Sobrevivieron / porcentaje total')
```



[21]:
```
# Gráfica de sexo y survived en cuenta total
data.Sex[data.Survived == 1].value_counts().plot(kind = 'barh', color =␣
 ↪"purple", alpha = 0.7)
plt.title("sobrevivieron totales - M vs F")
```

[21]: Text(0.5, 1.0, 'sobrevivieron totales - M vs F')



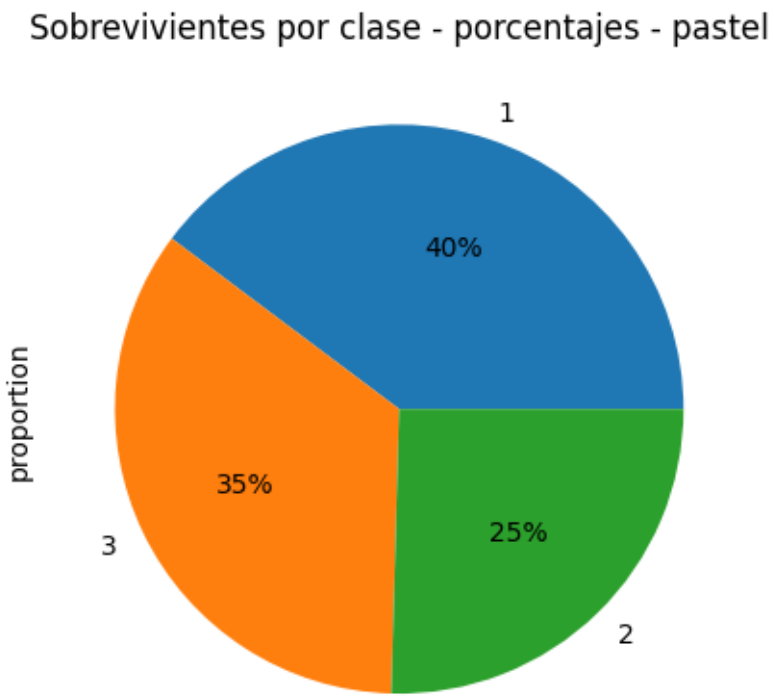
[22]:
```
# Gráfica de supervivencia por clase
data.Pclass[data.Survived == 1].value_counts().plot(kind = 'bar', color =␣
↪"green")
plt.title("Sobrevivientes por clase - Totales")
```

[22]: Text(0.5, 1.0, 'Sobrevivientes por clase - Totales')



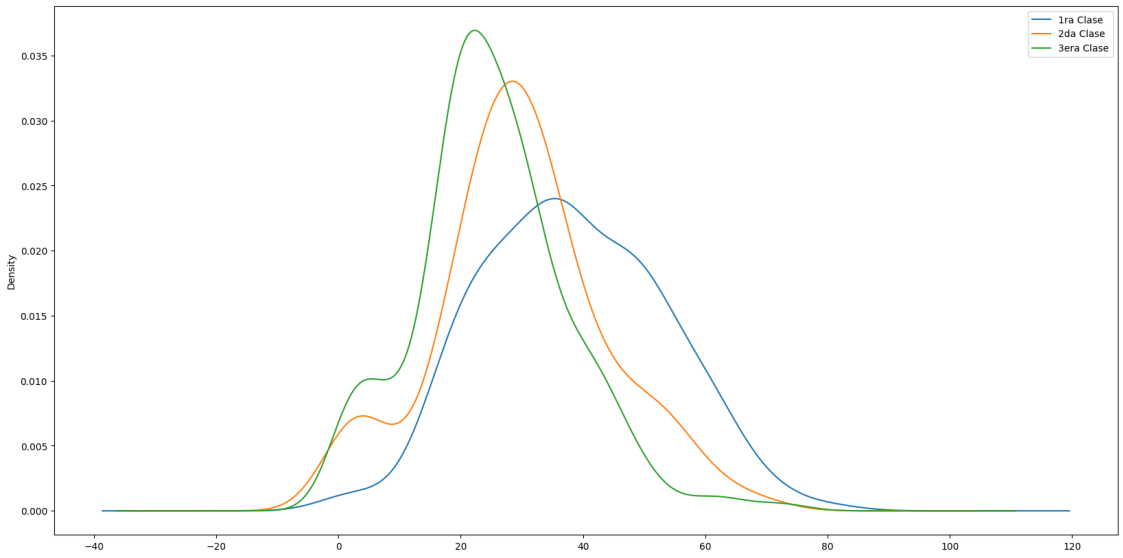
[23]:
```
# Gráfica de supervivencia por clase en porcentajes - pastel
data.Pclass[data.Survived == 1].value_counts(normalize = True).plot(kind =␣
 ↪"pie", autopct = "%1.0f%%")
plt.title("Sobrevivientes por clase - porcentajes - pastel")
```

[23]:
```
Text(0.5, 1.0, 'Sobrevivientes por clase - porcentajes - pastel')
```



[26]:
```
# Gráfica para ver si hay una relación entre tipo de ticket y edad
fig = plt.figure(figsize = (20,10))
for clase in [1,2,3]:
    data.Age[data.Pclass == clase].plot(kind = "kde")
plt.legend(("1ra Clase","2da Clase","3era Clase"))
```

[26]: <matplotlib.legend.Legend at 0x138a8ed60>



[32]:
```
# Información de la persona que pagó más para subirse
data[data.Fare == data.Fare.max()]
```

[32]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Age_months
258	259	1	1	Ward, Miss. Anna	F	35.0	0	0	PC 17755	512.3292	NaN	C	420
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	M	36.0	0	1	PC 17755	512.3292	B51 B53 B55	C	432
737	738	1	1	Lesurer, Mr. Gustave J	M	35.0	0	0	PC 17755	512.3292	B101	C	420

[28]:
```
# Cuantos pasajeros eran menores de 1 año
data[data.Age < 1]
```

[28]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
78	79	1	2	Caldwell, Master. Alden Gates	M	0.83	0	2	248738	29.0000	NaN	S
305	306	1	1	Allison, Master. Hudson Trevor	M	0.92	1	2	113781	151.5500	C22 C26	S
469	470	1	3	Baclini, Miss. Helene Barbara	F	0.75	2	1	2666	19.2583	NaN	C
644	645	1	3	Baclini, Miss. Eugenie	F	0.75						
755	756	1	2	Hamalainen, Master. Viljo	M	0.67						
803	804	1	3	Thomas, Master. Assad Alexander	M	0.42						
831	832	1	2	Richards, Master. George Sibley	M	0.83						

```
644      2      1    2666   19.2583      NaN        C
755      1      1  250649   14.5000      NaN        S
803      0      1    2625    8.5167      NaN        C
831      1      1   29106   18.7500      NaN        S
```

[29]:
```
# Columna de edad en meses
data["Age_months"] = (data["Age"] * 12).round().astype("Int64")
data.head()
```

[29]:
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name Sex   Age  SibSp  Parch  \
0                            Braund, Mr. Owen Harris   M  22.0      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th…   F  38.0      1      0
2                             Heikkinen, Miss. Laina   F  26.0      0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)   F  35.0      1      0
4                           Allen, Mr. William Henry   M  35.0      0      0

             Ticket     Fare Cabin Embarked  Age_months
0         A/5 21171   7.2500   NaN        S         264
1          PC 17599  71.2833   C85        C         456
2  STON/O2. 3101282   7.9250   NaN        S         312
3            113803  53.1000  C123        S         420
4            373450   8.0500   NaN        S         420
```

[31]:
```
data[data.Age<1]
```

[31]:
```
     PassengerId  Survived  Pclass                                Name Sex   Age  \
78            79         1       2   Caldwell, Master. Alden Gates   M  0.83
305          306         1       1  Allison, Master. Hudson Trevor   M  0.92
469          470         1       3   Baclini, Miss. Helene Barbara   F  0.75
644          645         1       3           Baclini, Miss. Eugenie   F  0.75
755          756         1       2      Hamalainen, Master. Viljo   M  0.67
803          804         1       3  Thomas, Master. Assad Alexander   M  0.42
831          832         1       2  Richards, Master. George Sibley   M  0.83

     SibSp  Parch  Ticket      Fare    Cabin Embarked  Age_months
78       0      2  248738   29.0000      NaN        S          10
305      1      2  113781  151.5500  C22 C26        S          11
469      2      1    2666   19.2583      NaN        C           9
644      2      1    2666   19.2583      NaN        C           9
755      1      1  250649   14.5000      NaN        S           8
```

```
803      0      1    2625    8.5167      NaN        C           5
831      1      1   29106   18.7500      NaN        S          10
```

[ ]: